

Advanced Database

Dr. M. Ali Nematbakhsh



Project1: Complex Data Types

1: Modeling and Analyzing JSON Data in Modern Databases

Suppose you are designing a database to store user information for a social network. Each user has attributes such as ID, Name, Age, City, Gender, and you can add additional Data at your choice.

- Modeling: Explain why using a JSON-based data model could be beneficial in this scenario
- Analysis: Discuss 3 advantages and 2 drawbacks of storing semi-structured Data in JSON format compared to a fully relational schema
- Sample Data: Provide a sample JSON Structure for 2 example users that illustrate the diversity of attributes
- Querying: write 5 different Queries based on your JSON files and explain how the Query structure is. you can use this platform: <https://jsonpath.com/>
- use JSON Query Engine to describe the past 5 queries that you done using Natural Language. for using API you dont need to use OpenAI, you can use free OpenRouter APIs instead. Here is the link that can help you to do this section: https://developers.llamaindex.ai/python/examples/query_engine/json_query_engine/

2: Comprehensive Comparison of XML and JSON

Consider the following scenario :

Complete information about a book, including its title, authors (with details such as name and nationality), publication year, publisher, ISBN, a list of chapters (each with title and starting page), and reader reviews (each with username, rating, and comment text)

- Model the above data using XML and JSON structures, and provide a complete data sample for each representation that clearly shows nested elements or objects and lists.

Compare XML and JSON from the following perspective :

- Readability: Which is more human-readable? why?
- Verbosity: Which consumes more storage space? Provide a brief numerical example comparing approximate file sizes
- Explain Different use cases that use JSON or XML

3: RDD and Knowledge Graph construction

Consider the following scenario :

An e-commerce company receives its product catalog in XML format, containing product_id, name, description, price, category, and customer reviews as a stream of JSON objects containing review_id, product_id, customer_name, rating, comment, and timestamp.

- Explain how the RDF can serve as a flexible and common data model to integrate these 2 heterogeneous sources (XML and JSON)
- Propose a simple schema for this knowledge graph. You must define the main classes (e.g., :Product, :Review, :Customer). The properties that link classes or assign literal values (e.g., :hasReview, :reviewedBy, :hasRating, :productName)

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What to upload for us?

Complete Documentation (PDF) + JSON files + XML files + All the sources you used

all your files must be in one ZIP file named as :

DBHW{Project Number}-{Student Complete name}-{Student Number}

the Template structure must be followed by all students. Failure to follow it will result in a grade deduction.

Dead Line : 5 November - 14 Aban

Behnam Soufi
Ali Moeinian